Maui stress-test annex

Data as of 2026-10-28

Planning aid. Supports, does not replace, county emergency plans. Not an official shelter list.

Stress test, not a forecast. Each test changes one thing in the plan (a site closes, two hazards strike together, a road is cut, more people come) and counts who is affected, from the plan's own data. Deterministic code computes every number from the plan's data; people review the results.

People counted: residents of Census areas whose drive from the area's centre point to the nearest usable 2026 site was 15 minutes or less before the loss and is more than 15 minutes (or no route) after it; areas already beyond 15 minutes are not counted.

174 tests ran for Maui; 115 affect someone or need county data and are listed below. Two-site losses are listed only when they add people beyond either site alone. This is not a statement that the area has no shelter: counties designate and open shelters for each event, and many public buildings are not on the 2026 list.

Hurricane: top 10

1. **Future sea-level-rise scenario (3.2 ft; State of Hawai‘i SLR-XA), tested against 2026 sites – Maui: every potentially flooded highway segment closed** (test 24: Road closures (driving))
   70,875 people in areas with a longer drive, fewer known sites within 45 minutes, or cut off (+92 min to any known site; +3.8 min to a 2026 site; cut off from every 2026 site: Launiupoko / Mā‘alaea · Census area 320).

   Census areas cut off: Launiupoko / Mā‘alaea · Census area 320 · Census areas cut off (over the limit or no route): 1 · Census areas cut off from every 2026 site: 1 · Census areas losing known sites within reach: 18 · Census areas with no route at all: 1 · Census areas routed: 27 · Census areas with a longer drive to a 2026 site: 3 · Cut off means: over 45 minutes or no route · Most minutes added to any known site: 92 · Most minutes added to the nearest 2026 site: 3.8 · Typical minutes added (median, where longer): 2.4 · People in areas cut off: 951 · People with a longer trip: 12,064 · People losing known sites within reach: 68,353 · Within reach means: 45-minute drive · Why this road:

   Sources: Esri ArcGIS routing with road closures (build time); highway centrelines: Hawai‘i DOT HPMS and surveyed mile markers (Hawai‘i Statewide GIS); Esri ArcGIS routing (drive-time matrix, normal traffic, build time); county announcements of sites opened in 2026; FEMA flood zones and tsunami evacuation zones (Hawai‘i Statewide GIS); Hawai‘i DOE public schools (Hawai‘i Statewide GIS); Hawai‘i Sea Level Rise Viewer, potential flooded highways at 3.2 ft (PacIOOS, via Hawai‘i Statewide GIS). Team assumptions: Future sea-level-rise scenario (3.2 ft; State of Hawai‘i SLR-XA), tested against 2026 sites: every potentially flooded highway segment on the island closed at once: all segments.

2. **Lose Baldwin High School and Mayor Hannibal Tavares Community Center (Social Hall) (hurricane)** (test 2: Two sites lost)
   69,957 people in the risk area pushed outside a 15-minute drive to a usable 2026 site (Census areas pushed outside a 15-minute drive: 15; people more than losing either site alone: 23,964).

   Census areas left with no usable 2026 site on the island: 0 · Census areas pushed outside a 15-minute drive: 15 · Drive limit (minutes): 15 · Island: Maui · People more than losing either site alone: 23,964 · People left with no usable 2026 site on the island: 0 · Sites: Baldwin High School, Mayor Hannibal Tavares Community Center (Social Hall)

   Sources: CDC/ATSDR Social Vulnerability Index 2022 (ACS 2018–2022), HHS emPOWER, FEMA National Risk Index; 2020 Census blocks; county announcements of sites opened in 2026; FEMA flood zones and tsunami evacuation zones (Hawai‘i Statewide GIS); Esri ArcGIS routing (drive-time matrix, normal traffic, build time).

3. **Lose Baldwin High School and South Maui Community Park Gym (hurricane)** (test 2: Two sites lost)
   68,514 people in the risk area pushed outside a 15-minute drive to a usable 2026 site (Census areas pushed outside a 15-minute drive: 18; people more than losing either site alone: 22,521).
   Census areas left with no usable 2026 site on the island: 0 · Census areas pushed outside a 15-minute drive: 18 · Drive limit (minutes): 15 · Island: Maui · People more than losing either site alone: 22,521 · People left with no usable 2026 site on the island: 0 · Sites: Baldwin High School, South Maui Community Park Gym
   Sources: CDC/ATSDR Social Vulnerability Index 2022 (ACS 2018–2022), HHS emPOWER, FEMA National Risk Index; 2020 Census blocks; county announcements of sites opened in 2026; FEMA flood zones and tsunami evacuation zones (Hawai‘i Statewide GIS); Esri ArcGIS routing (drive-time matrix, normal traffic, build time).
4. **Honoapi‘ilani Highway (Route 30) at the Pali and at Olowalu** (test 24: Road closures (driving))
   68,353 people in areas with a longer drive, fewer known sites within 45 minutes, or cut off (+103.7 min to any known site; +0 min to a 2026 site).
   Census areas cut off: none · Census areas cut off (over the limit or no route): 0 · Census areas cut off from every 2026 site: 0 · Census areas losing known sites within reach: 18 · Census areas with no route at all: 0 · Census areas routed: 41 · Census areas with a longer drive to a 2026 site: 0 · Cut off means: over 45 minutes or no route · Most minutes added to any known site: 103.7 · Most minutes added to the nearest 2026 site: 0 · Typical minutes added (median, where longer): 0 · People in areas cut off: 0 · People with a longer trip: 0 · People losing known sites within reach: 68,353 · Within reach means: 45-minute drive · Why this road: Only coastal highway between West Maui and Central Maui; the Pali stretch is exposed to rockfall and the Olowalu stretch to wildfire and coastal flooding (team assumption).
   Sources: Esri ArcGIS routing with road closures (build time); highway centrelines: Hawai‘i DOT HPMS and surveyed mile markers (Hawai‘i Statewide GIS); Esri ArcGIS routing (drive-time matrix, normal traffic, build time); county announcements of sites opened in 2026; FEMA flood zones and tsunami evacuation zones (Hawai‘i Statewide GIS); Hawai‘i DOE public schools (Hawai‘i Statewide GIS). Team assumptions: Closure location along the highway (a stretch of about 0.8 km): team assumption.
5. **Stranded visitors for 3 or 7 days – Maui** (test 21: Stranded visitors)
   55,290 visitors on an average day (island level).
   By length of stay: 3 days: 331,740 meals, 165,870 gallons of water; 7 days: 774,060 meals, 387,030 gallons of water · Visitors on an average day (2025): 55,290
   Sources: DBEDT State of Hawai‘i Data Book 2025, Table 7.06 (average daily visitor census, 2025); Florida Division of Emergency Management, Sheltering Guidance Aid (2010). Team assumptions: Every visitor on the island is stranded and needs meals and water from mass care (upper bound): 100%.
6. **Lose Baldwin High School (hurricane)** (test 1: One site lost)
   45,993 people in the risk area pushed outside a 15-minute drive to a usable 2026 site (Census areas pushed outside a 15-minute drive: 10).
   Census areas pushed outside a 15-minute drive: 10 · Drive limit (minutes): 15 · Island: Maui · May need a public shelter (likely): 1,380 · Site: Baldwin High School
   Sources: CDC/ATSDR Social Vulnerability Index 2022 (ACS 2018–2022), HHS emPOWER, FEMA National Risk Index; 2020 Census blocks; county announcements of sites opened in 2026; FEMA flood zones and tsunami evacuation zones (Hawai‘i Statewide GIS); Esri ArcGIS routing (drive-time matrix, normal traffic, build time).

7. **All four counties at once (hurricane)** (test 9: Several counties at once)
   43,518 people who may need a public shelter (likely), counties together (cots: 43,518; shelter staff per shift: 2,921).
   Cots: 43,518 · Counties: Kauai, Honolulu, Maui, Hawaii · People with limited English in shelter, by language (interpreters): Chinese: 247; Chuukese: 58; Hawaiian: 28; Ilocano: 369; Japanese: 201; Korean: 99; Marshallese: 24; Pidgin (Hawai‘i Creole English): 25; Spanish: 29; Tagalog/Filipino: 303; Vietnamese: 66 · Meals per day: 87,036 · People in shelter (likely): 43,518 · Rounding: counties' unrounded numbers added, then rounded up once · Shelter staff per shift: 2,921 · Water (gallons per day): 43,518
   Sources: CDC/ATSDR Social Vulnerability Index 2022 (ACS 2018–2022), HHS emPOWER, FEMA National Risk Index; 2020 Census blocks; Florida Division of Emergency Management, Sheltering Guidance Aid (2010); ACS PUMS 2019–2023 5-year (limited-English speakers by language); shelter-seeking rates in the plan (South Carolina EMD 2026 Annex G; Maui 2023 observed; team assumptions). Team assumptions: One rounding rule for counties together: add the unrounded county numbers, then round up once: round up.

8. **Honolulu and Maui at once (hurricane)** (test 9: Several counties at once)
   35,248 people who may need a public shelter (likely), counties together (cots: 35,248; shelter staff per shift: 2,357).
   Cots: 35,248 · Counties: Honolulu, Maui · People with limited English in shelter, by language (interpreters): Chinese: 240; Chuukese: 58; Hawaiian: 19; Ilocano: 302; Japanese: 181; Korean: 99; Marshallese: 16; Spanish: 10; Tagalog/Filipino: 278; Vietnamese: 66 · Meals per day: 70,495 · People in shelter (likely): 35,248 · Rounding: counties' unrounded numbers added, then rounded up once · Shelter staff per shift: 2,357 · Water (gallons per day): 35,248
   Sources: CDC/ATSDR Social Vulnerability Index 2022 (ACS 2018–2022), HHS emPOWER, FEMA National Risk Index; 2020 Census blocks; Florida Division of Emergency Management, Sheltering Guidance Aid (2010); ACS PUMS 2019–2023 5-year (limited-English speakers by language); shelter-seeking rates in the plan (South Carolina EMD 2026 Annex G; Maui 2023 observed; team assumptions). Team assumptions: One rounding rule for counties together: add the unrounded county numbers, then round up once: round up.

9. **Lose South Maui Community Park Gym (hurricane)** (test 1: One site lost)
   22,521 people in the risk area pushed outside a 15-minute drive to a usable 2026 site (Census areas pushed outside a 15-minute drive: 8).
   Census areas pushed outside a 15-minute drive: 8 · Drive limit (minutes): 15 · Island: Maui · May need a public shelter (likely): 676 · Site: South Maui Community Park Gym
   Sources: CDC/ATSDR Social Vulnerability Index 2022 (ACS 2018–2022), HHS emPOWER, FEMA National Risk Index; 2020 Census blocks; county announcements of sites opened in 2026; FEMA flood zones and tsunami evacuation zones (Hawai‘i Statewide GIS); Esri ArcGIS routing (drive-time matrix, normal traffic, build time).

10. **Honoapiʻilani Highway (Route 30) at the Pali and at Olowalu – walking** (test 25: Road closures (walking))

    12,413 people in areas with a longer walk to a 2026 site or cut off on foot (+627.6 min to any known site; +627.6 min to a 2026 site).

    Census areas cut off: none · Census areas cut off (over the limit or no route): 0 · Census areas cut off from every 2026 site: 0 · Census areas losing known sites within reach: 0 · Census areas with no route at all: 0 · Census areas routed: 39 · Census areas with a longer drive to a 2026 site: 3 · Cut off means: no walking route · Most minutes added to any known site: 627.6 · Most minutes added to the nearest 2026 site: 627.6 · Typical minutes added (median, where longer): 12.9 · People in areas cut off: 0 · People with a longer trip: 12,413 · People losing known sites within reach: 0 · Within reach means: walking route (sites within 12 km) · Why this road: Only coastal highway between West Maui and Central Maui; the Pali stretch is exposed to rockfall and the Olowalu stretch to wildfire and coastal flooding (team assumption).

    Sources: Esri ArcGIS routing with road closures (build time); highway centrelines: Hawaiʻi DOT HPMS and surveyed mile markers (Hawaiʻi Statewide GIS); Esri ArcGIS routing (walking time, build time); county announcements of sites opened in 2026; FEMA flood zones and tsunami evacuation zones (Hawaiʻi Statewide GIS); Hawaiʻi DOE public schools (Hawaiʻi Statewide GIS). Team assumptions: Closure location along the highway (a stretch of about 0.8 km): team assumption.

Wildfire: top 10

1. **Future sea-level-rise scenario (3.2 ft; State of Hawai‘i SLR-XA), tested against 2026 sites – Maui: every potentially flooded highway segment closed** (test 24: Road closures (driving))
   70,875 people in areas with a longer drive, fewer known sites within 45 minutes, or cut off (+92 min to any known site; +3.8 min to a 2026 site; cut off from every 2026 site: Launiupoko / Mā‘alaea · Census area 320).

   Census areas cut off: Launiupoko / Mā‘alaea · Census area 320 · Census areas cut off (over the limit or no route): 1 · Census areas cut off from every 2026 site: 1 · Census areas losing known sites within reach: 18 · Census areas with no route at all: 1 · Census areas routed: 27 · Census areas with a longer drive to a 2026 site: 3 · Cut off means: over 45 minutes or no route · Most minutes added to any known site: 92 · Most minutes added to the nearest 2026 site: 3.8 · Typical minutes added (median, where longer): 2.4 · People in areas cut off: 951 · People with a longer trip: 12,064 · People losing known sites within reach: 68,353 · Within reach means: 45-minute drive · Why this road:

   Sources: Esri ArcGIS routing with road closures (build time); highway centrelines: Hawai‘i DOT HPMS and surveyed mile markers (Hawai‘i Statewide GIS); Esri ArcGIS routing (drive-time matrix, normal traffic, build time); county announcements of sites opened in 2026; FEMA flood zones and tsunami evacuation zones (Hawai‘i Statewide GIS); Hawai‘i DOE public schools (Hawai‘i Statewide GIS); Hawai‘i Sea Level Rise Viewer, potential flooded highways at 3.2 ft (PacIOOS, via Hawai‘i Statewide GIS). Team assumptions: Future sea-level-rise scenario (3.2 ft; State of Hawai‘i SLR-XA), tested against 2026 sites: every potentially flooded highway segment on the island closed at once: all segments.

2. **Honoapi‘ilani Highway (Route 30) at the Pali and at Olowalu** (test 24: Road closures (driving))
   68,353 people in areas with a longer drive, fewer known sites within 45 minutes, or cut off (+103.7 min to any known site; +0 min to a 2026 site).

   Census areas cut off: none · Census areas cut off (over the limit or no route): 0 · Census areas cut off from every 2026 site: 0 · Census areas losing known sites within reach: 18 · Census areas with no route at all: 0 · Census areas routed: 41 · Census areas with a longer drive to a 2026 site: 0 · Cut off means: over 45 minutes or no route · Most minutes added to any known site: 103.7 · Most minutes added to the nearest 2026 site: 0 · Typical minutes added (median, where longer): 0 · People in areas cut off: 0 · People with a longer trip: 0 · People losing known sites within reach: 68,353 · Within reach means: 45-minute drive · Why this road: Only coastal highway between West Maui and Central Maui; the Pali stretch is exposed to rockfall and the Olowalu stretch to wildfire and coastal flooding (team assumption).

   Sources: Esri ArcGIS routing with road closures (build time); highway centrelines: Hawai‘i DOT HPMS and surveyed mile markers (Hawai‘i Statewide GIS); Esri ArcGIS routing (drive-time matrix, normal traffic, build time); county announcements of sites opened in 2026; FEMA flood zones and tsunami evacuation zones (Hawai‘i Statewide GIS); Hawai‘i DOE public schools (Hawai‘i Statewide GIS). Team assumptions: Closure location along the highway (a stretch of about 0.8 km): team assumption.

3. **Areas high in the State Fire Risk layer but not in the plan's wildfire areas, and the reverse – Maui** (test 23b: State Fire Risk Areas vs the plan's wildfire areas)
   56,637 people in areas rated high only by the State Fire Risk layer (Census areas high in the State layer only: 18; Census areas in the plan's wildfire areas only: 4).
   Areas high in both, of all areas high in either: 27% · Census areas high in both: 8 · Census areas in the plan's wildfire areas only: 4 · Census areas high in the State layer only: 18 · Largest plan-only areas: Wailuku · Census area 310.01, Kula · Census area 303.05, Kula / Kēōkea · Census area 303.04, Hāna · Census area 301 · People in plan-only areas: 17,216 · People in State-only areas: 56,637 · People in the plan's wildfire areas (FEMA National Risk Index): 48,834 · People in State High fire risk areas: 87,832 · Largest State-only areas: Kahului · Census area 311.05, Kahului · Census area 311.02, Kahului · Census area 311.06, Kīhei · Census area 307.13, Wailuku · Census area 310.02
   Sources: State Fire Risk Areas (DLNR Division of Forestry and Wildlife, via Hawai‘i Statewide GIS); FEMA National Risk Index wildfire scores; CDC/ATSDR Social Vulnerability Index 2022 (ACS 2018–2022), HHS emPOWER, FEMA National Risk Index; 2020 Census blocks. Team assumptions: An area is 'high' in the State layer when at least half its people live in a High fire risk area: 50%.
4. **Stranded visitors for 3 or 7 days – Maui** (test 21: Stranded visitors)
   55,290 visitors on an average day (island level).
   By length of stay: 3 days: 331,740 meals, 165,870 gallons of water; 7 days: 774,060 meals, 387,030 gallons of water · Visitors on an average day (2025): 55,290
   Sources: DBEDT State of Hawai‘i Data Book 2025, Table 7.06 (average daily visitor census, 2025); Florida Division of Emergency Management, Sheltering Guidance Aid (2010). Team assumptions: Every visitor on the island is stranded and needs meals and water from mass care (upper bound): 100%.
5. **All four counties at once (wildfire)** (test 9: Several counties at once)
   22,115 people who may need a public shelter (likely), counties together (cots: 22,115; shelter staff per shift: 1,534).
   Cots: 22,115 · Counties: Kauai, Honolulu, Maui, Hawaii · People with limited English in shelter, by language (interpreters): Chinese: 61; Chuukese: 14; Hawaiian: 9; Ilocano: 110; Japanese: 54; Korean: 24; Marshallese: 8; Pidgin (Hawai‘i Creole English): 10; Spanish: 12; Tagalog/Filipino: 84; Vietnamese: 16 · Meals per day: 44,229 · People in shelter (likely): 22,115 · Rounding: counties' unrounded numbers added, then rounded up once · Shelter staff per shift: 1,534 · Water (gallons per day): 22,115
   Sources: CDC/ATSDR Social Vulnerability Index 2022 (ACS 2018–2022), HHS emPOWER, FEMA National Risk Index; 2020 Census blocks; Florida Division of Emergency Management, Sheltering Guidance Aid (2010); ACS PUMS 2019–2023 5-year (limited-English speakers by language); shelter-seeking rates in the plan (South Carolina EMD 2026 Annex G; Maui 2023 observed; team assumptions). Team assumptions: One rounding rule for counties together: add the unrounded county numbers, then round up once: round up.

6. **Honolulu and Maui at once (wildfire)** (test 9: Several counties at once)

   18,155 people who may need a public shelter (likely), counties together (cots: 18,155; shelter staff per shift: 1,258).

   Cots: 18,155 · Counties: Honolulu, Maui · People with limited English in shelter, by language (interpreters): Chinese: 58; Chuukese: 14; Hawaiian: 6; Ilocano: 91; Japanese: 47; Korean: 24; Marshallese: 5; Spanish: 6; Tagalog/Filipino: 77; Vietnamese: 16 · Meals per day: 36,310 · People in shelter (likely): 18,155 · Rounding: counties' unrounded numbers added, then rounded up once · Shelter staff per shift: 1,258 · Water (gallons per day): 18,155

   Sources: CDC/ATSDR Social Vulnerability Index 2022 (ACS 2018–2022), HHS emPOWER, FEMA National Risk Index; 2020 Census blocks; Florida Division of Emergency Management, Sheltering Guidance Aid (2010); ACS PUMS 2019–2023 5-year (limited-English speakers by language); shelter-seeking rates in the plan (South Carolina EMD 2026 Annex G; Maui 2023 observed; team assumptions). Team assumptions: One rounding rule for counties together: add the unrounded county numbers, then round up once: round up.

7. **Honoapi‘ilani Highway (Route 30) at the Pali and at Olowalu – walking** (test 25: Road closures (walking))

   12,413 people in areas with a longer walk to a 2026 site or cut off on foot (+627.6 min to any known site; +627.6 min to a 2026 site).

   Census areas cut off: none · Census areas cut off (over the limit or no route): 0 · Census areas cut off from every 2026 site: 0 · Census areas losing known sites within reach: 0 · Census areas with no route at all: 0 · Census areas routed: 39 · Census areas with a longer drive to a 2026 site: 3 · Cut off means: no walking route · Most minutes added to any known site: 627.6 · Most minutes added to the nearest 2026 site: 627.6 · Typical minutes added (median, where longer): 12.9 · People in areas cut off: 0 · People with a longer trip: 12,413 · People losing known sites within reach: 0 · Within reach means: walking route (sites within 12 km) · Why this road: Only coastal highway between West Maui and Central Maui; the Pali stretch is exposed to rockfall and the Olowalu stretch to wildfire and coastal flooding (team assumption).

   Sources: Esri ArcGIS routing with road closures (build time); highway centrelines: Hawai‘i DOT HPMS and surveyed mile markers (Hawai‘i Statewide GIS); Esri ArcGIS routing (walking time, build time); county announcements of sites opened in 2026; FEMA flood zones and tsunami evacuation zones (Hawai‘i Statewide GIS); Hawai‘i DOE public schools (Hawai‘i Statewide GIS). Team assumptions: Closure location along the highway (a stretch of about 0.8 km): team assumption.

8. **Lose Baldwin High School and Mayor Hannibal Tavares Community Center (Social Hall) (wildfire)** (test 2: Two sites lost)

   12,302 people in the risk area pushed outside a 15-minute drive to a usable 2026 site (Census areas pushed outside a 15-minute drive: 2; people more than losing either site alone: 5,325).

   Census areas left with no usable 2026 site on the island: 0 · Census areas pushed outside a 15-minute drive: 2 · Drive limit (minutes): 15 · Island: Maui · People more than losing either site alone: 5,325 · People left with no usable 2026 site on the island: 0 · Sites: Baldwin High School, Mayor Hannibal Tavares Community Center (Social Hall)

   Sources: CDC/ATSDR Social Vulnerability Index 2022 (ACS 2018–2022), HHS emPOWER, FEMA National Risk Index; 2020 Census blocks; county announcements of sites opened in 2026; FEMA flood zones and tsunami evacuation zones (Hawai‘i Statewide GIS); Esri ArcGIS routing (drive-time matrix, normal traffic, build time).

9. **Hāna Highway (Route 360) near Keʻanae – walking** (test 25: Road closures (walking))
   11,462 people in areas with a longer walk to a 2026 site or cut off on foot (+12.9 min to any known site; +12.9 min to a 2026 site).
   Census areas cut off: none · Census areas cut off (over the limit or no route): 0 · Census areas cut off from every 2026 site: 0 · Census areas losing known sites within reach: 0 · Census areas with no route at all: 0 · Census areas routed: 39 · Census areas with a longer drive to a 2026 site: 2 · Cut off means: no walking route · Most minutes added to any known site: 12.9 · Most minutes added to the nearest 2026 site: 12.9 · Typical minutes added (median, where longer): 12.6 · People in areas cut off: 0 · People with a longer trip: 11,462 · People losing known sites within reach: 0 · Within reach means: walking route (sites within 12 km) · Why this road: Narrow road with many one-lane bridges and landslide-prone cliffs (team assumption).
   Sources: Esri ArcGIS routing with road closures (build time); highway centrelines: Hawai‘i DOT HPMS and surveyed mile markers (Hawai‘i Statewide GIS); Esri ArcGIS routing (walking time, build time); county announcements of sites opened in 2026; FEMA flood zones and tsunami evacuation zones (Hawai‘i Statewide GIS); Hawai‘i DOE public schools (Hawai‘i Statewide GIS). Team assumptions: Closure location along the highway (a stretch of about 0.8 km): team assumption.

10. **Lose Baldwin High School and South Maui Community Park Gym (wildfire)** (test 2: Two sites lost)
    10,561 people in the risk area pushed outside a 15-minute drive to a usable 2026 site (Census areas pushed outside a 15-minute drive: 2; people more than losing either site alone: 3,584).
    Census areas left with no usable 2026 site on the island: 0 · Census areas pushed outside a 15-minute drive: 2 · Drive limit (minutes): 15 · Island: Maui · People more than losing either site alone: 3,584 · People left with no usable 2026 site on the island: 0 · Sites: Baldwin High School, South Maui Community Park Gym
    Sources: CDC/ATSDR Social Vulnerability Index 2022 (ACS 2018–2022), HHS emPOWER, FEMA National Risk Index; 2020 Census blocks; county announcements of sites opened in 2026; FEMA flood zones and tsunami evacuation zones (Hawai‘i Statewide GIS); Esri ArcGIS routing (drive-time matrix, normal traffic, build time).

Tsunami: top 10

1. **Future sea-level-rise scenario (3.2 ft; State of Hawai‘i SLR-XA), tested against 2026 sites – Maui: every potentially flooded highway segment closed** (test 24: Road closures (driving))
   70,875 people in areas with a longer drive, fewer known sites within 45 minutes, or cut off (+92 min to any known site; +3.8 min to a 2026 site; cut off from every 2026 site: Launiupoko / Mā‘alaea · Census area 320).

   Census areas cut off: Launiupoko / Mā‘alaea · Census area 320 · Census areas cut off (over the limit or no route): 1 · Census areas cut off from every 2026 site: 1 · Census areas losing known sites within reach: 18 · Census areas with no route at all: 1 · Census areas routed: 27 · Census areas with a longer drive to a 2026 site: 3 · Cut off means: over 45 minutes or no route · Most minutes added to any known site: 92 · Most minutes added to the nearest 2026 site: 3.8 · Typical minutes added (median, where longer): 2.4 · People in areas cut off: 951 · People with a longer trip: 12,064 · People losing known sites within reach: 68,353 · Within reach means: 45-minute drive · Why this road:

   Sources: Esri ArcGIS routing with road closures (build time); highway centrelines: Hawai‘i DOT HPMS and surveyed mile markers (Hawai‘i Statewide GIS); Esri ArcGIS routing (drive-time matrix, normal traffic, build time); county announcements of sites opened in 2026; FEMA flood zones and tsunami evacuation zones (Hawai‘i Statewide GIS); Hawai‘i DOE public schools (Hawai‘i Statewide GIS); Hawai‘i Sea Level Rise Viewer, potential flooded highways at 3.2 ft (PacIOOS, via Hawai‘i Statewide GIS). Team assumptions: Future sea-level-rise scenario (3.2 ft; State of Hawai‘i SLR-XA), tested against 2026 sites: every potentially flooded highway segment on the island closed at once: all segments.

2. **Honoapi‘ilani Highway (Route 30) at the Pali and at Olowalu** (test 24: Road closures (driving))
   68,353 people in areas with a longer drive, fewer known sites within 45 minutes, or cut off (+103.7 min to any known site; +0 min to a 2026 site).

   Census areas cut off: none · Census areas cut off (over the limit or no route): 0 · Census areas cut off from every 2026 site: 0 · Census areas losing known sites within reach: 18 · Census areas with no route at all: 0 · Census areas routed: 41 · Census areas with a longer drive to a 2026 site: 0 · Cut off means: over 45 minutes or no route · Most minutes added to any known site: 103.7 · Most minutes added to the nearest 2026 site: 0 · Typical minutes added (median, where longer): 0 · People in areas cut off: 0 · People with a longer trip: 0 · People losing known sites within reach: 68,353 · Within reach means: 45-minute drive · Why this road: Only coastal highway between West Maui and Central Maui; the Pali stretch is exposed to rockfall and the Olowalu stretch to wildfire and coastal flooding (team assumption).

   Sources: Esri ArcGIS routing with road closures (build time); highway centrelines: Hawai‘i DOT HPMS and surveyed mile markers (Hawai‘i Statewide GIS); Esri ArcGIS routing (drive-time matrix, normal traffic, build time); county announcements of sites opened in 2026; FEMA flood zones and tsunami evacuation zones (Hawai‘i Statewide GIS); Hawai‘i DOE public schools (Hawai‘i Statewide GIS). Team assumptions: Closure location along the highway (a stretch of about 0.8 km): team assumption.

3. **Visitors on the island(s) – Maui (tsunami)** (test 20: Visitors in tsunami zones)
   55,290 visitors on an average day (island level) (visitors inside the tsunami zone: needs county data (hotel and rental locations)).
   Detail: Maui 54,500; Molokaʻi 385; Lānaʻi 406 · Peak day: needs DBEDT monthly data (the Data Book gives the annual average) · Visitors on an average day (2025): 55,290 · Visitors inside the tsunami zone: needs county data (hotel and rental locations)
   Sources: DBEDT State of Hawaiʻi Data Book 2025, Table 7.06 (average daily visitor census, 2025).
4. **Stranded visitors for 3 or 7 days – Maui** (test 21: Stranded visitors)
   55,290 visitors on an average day (island level).
   By length of stay: 3 days: 331,740 meals, 165,870 gallons of water; 7 days: 774,060 meals, 387,030 gallons of water · Visitors on an average day (2025): 55,290
   Sources: DBEDT State of Hawaiʻi Data Book 2025, Table 7.06 (average daily visitor census, 2025); Florida Division of Emergency Management, Sheltering Guidance Aid (2010). Team assumptions: Every visitor on the island is stranded and needs meals and water from mass care (upper bound): 100%.
5. **Extreme tsunami zone (worst case) – Maui** (test 8: Extreme tsunami zone (worst case))
   38,773 people in the extreme tsunami evacuation zone (may need a public shelter (likely): 1,939; 2026 sites inside the extreme zone: 0).
   People in the extreme tsunami zone: 38,773 · People in the standard tsunami zone: 26,158 · May need a shelter (high): 4,265 · May need a public shelter (likely): 1,939 · May need a shelter (low): 775 · 2026 sites inside the extreme zone: 0 · Sites needed at 250 people per site (team assumption): 8 · 2026 sites usable outside both zones: 7
   Sources: CDC/ATSDR Social Vulnerability Index 2022 (ACS 2018–2022), HHS emPOWER, FEMA National Risk Index; 2020 Census blocks; State tsunami evacuation zones and extreme tsunami evacuation zones (Hawaiʻi Statewide GIS); county announcements of sites opened in 2026; FEMA flood zones and tsunami evacuation zones (Hawaiʻi Statewide GIS); shelter-seeking rates in the plan (South Carolina EMD 2026 Annex G; Maui 2023 observed; team assumptions). Team assumptions: Sites needed at a given number of people per site (capacity unknown): 250.
6. **Honoapiʻilani Highway (Route 30) at the Pali and at Olowalu – walking** (test 25: Road closures (walking))
   12,413 people in areas with a longer walk to a 2026 site or cut off on foot (+627.6 min to any known site; +627.6 min to a 2026 site).
   Census areas cut off: none · Census areas cut off (over the limit or no route): 0 · Census areas cut off from every 2026 site: 0 · Census areas losing known sites within reach: 0 · Census areas with no route at all: 0 · Census areas routed: 39 · Census areas with a longer drive to a 2026 site: 3 · Cut off means: no walking route · Most minutes added to any known site: 627.6 · Most minutes added to the nearest 2026 site: 627.6 · Typical minutes added (median, where longer): 12.9 · People in areas cut off: 0 · People with a longer trip: 12,413 · People losing known sites within reach: 0 · Within reach means: walking route (sites within 12 km) · Why this road: Only coastal highway between West Maui and Central Maui; the Pali stretch is exposed to rockfall and the Olowalu stretch to wildfire and coastal flooding (team assumption).
   Sources: Esri ArcGIS routing with road closures (build time); highway centrelines: Hawaiʻi DOT HPMS and surveyed mile markers (Hawaiʻi Statewide GIS); Esri ArcGIS routing (walking time, build time); county announcements of sites opened in 2026; FEMA flood zones and tsunami evacuation zones (Hawaiʻi Statewide GIS); Hawaiʻi DOE public schools (Hawaiʻi Statewide GIS). Team assumptions: Closure location along the highway (a stretch of about 0.8 km): team assumption.

7. **Lose Lahaina Civic Center Gym and South Maui Community Park Gym (tsunami)** (test 2: Two sites lost)
   12,189 people in the risk area pushed outside a 15-minute drive to a usable 2026 site (Census areas pushed outside a 15-minute drive: 8; people more than losing either site alone: 3,072).
   Census areas left with no usable 2026 site on the island: 0 · Census areas pushed outside a 15-minute drive: 8 · Drive limit (minutes): 15 · Island: Maui · People more than losing either site alone: 3,072 · People left with no usable 2026 site on the island: 0 · Sites: Lahaina Civic Center Gym, South Maui Community Park Gym
   Sources: CDC/ATSDR Social Vulnerability Index 2022 (ACS 2018–2022), HHS emPOWER, FEMA National Risk Index; 2020 Census blocks; county announcements of sites opened in 2026; FEMA flood zones and tsunami evacuation zones (Hawai‘i Statewide GIS); Esri ArcGIS routing (drive-time matrix, normal traffic, build time).

8. **Hāna Highway (Route 360) near Ke‘anae – walking** (test 25: Road closures (walking))
   11,462 people in areas with a longer walk to a 2026 site or cut off on foot (+12.9 min to any known site; +12.9 min to a 2026 site).
   Census areas cut off: none · Census areas cut off (over the limit or no route): 0 · Census areas cut off from every 2026 site: 0 · Census areas losing known sites within reach: 0 · Census areas with no route at all: 0 · Census areas routed: 39 · Census areas with a longer drive to a 2026 site: 2 · Cut off means: no walking route · Most minutes added to any known site: 12.9 · Most minutes added to the nearest 2026 site: 12.9 · Typical minutes added (median, where longer): 12.6 · People in areas cut off: 0 · People with a longer trip: 11,462 · People losing known sites within reach: 0 · Within reach means: walking route (sites within 12 km) · Why this road: Narrow road with many one-lane bridges and landslide-prone cliffs (team assumption).
   Sources: Esri ArcGIS routing with road closures (build time); highway centrelines: Hawai‘i DOT HPMS and surveyed mile markers (Hawai‘i Statewide GIS); Esri ArcGIS routing (walking time, build time); county announcements of sites opened in 2026; FEMA flood zones and tsunami evacuation zones (Hawai‘i Statewide GIS); Hawai‘i DOE public schools (Hawai‘i Statewide GIS). Team assumptions: Closure location along the highway (a stretch of about 0.8 km): team assumption.

9. **Lose Baldwin High School and South Maui Community Park Gym (tsunami)** (test 2: Two sites lost)
   11,323 people in the risk area pushed outside a 15-minute drive to a usable 2026 site (Census areas pushed outside a 15-minute drive: 10; people more than losing either site alone: 2,206).
   Census areas left with no usable 2026 site on the island: 0 · Census areas pushed outside a 15-minute drive: 10 · Drive limit (minutes): 15 · Island: Maui · People more than losing either site alone: 2,206 · People left with no usable 2026 site on the island: 0 · Sites: Baldwin High School, South Maui Community Park Gym
   Sources: CDC/ATSDR Social Vulnerability Index 2022 (ACS 2018–2022), HHS emPOWER, FEMA National Risk Index; 2020 Census blocks; county announcements of sites opened in 2026; FEMA flood zones and tsunami evacuation zones (Hawai‘i Statewide GIS); Esri ArcGIS routing (drive-time matrix, normal traffic, build time).

10. **Lose South Maui Community Park Gym (tsunami)** (test 1: One site lost)
    9,117 people in the risk area pushed outside a 15-minute drive to a usable 2026 site (Census areas pushed outside a 15-minute drive: 7).
    Census areas pushed outside a 15-minute drive: 7 · Drive limit (minutes): 15 · Island: Maui · May need a public shelter (likely): 456 · Site: South Maui Community Park Gym
    Sources: CDC/ATSDR Social Vulnerability Index 2022 (ACS 2018–2022), HHS emPOWER, FEMA National Risk Index; 2020 Census blocks; county announcements of sites opened in 2026; FEMA flood zones and tsunami evacuation zones (Hawai‘i Statewide GIS); Esri ArcGIS routing (drive-time matrix, normal traffic, build time).

Inland flooding: top 10

1. **Future sea-level-rise scenario (3.2 ft; State of Hawai‘i SLR-XA), tested against 2026 sites – Maui: every potentially flooded highway segment closed** (test 24: Road closures (driving))
   70,875 people in areas with a longer drive, fewer known sites within 45 minutes, or cut off (+92 min to any known site; +3.8 min to a 2026 site; cut off from every 2026 site: Launiupoko / Mā‘alaea · Census area 320).

   Census areas cut off: Launiupoko / Mā‘alaea · Census area 320 · Census areas cut off (over the limit or no route): 1 · Census areas cut off from every 2026 site: 1 · Census areas losing known sites within reach: 18 · Census areas with no route at all: 1 · Census areas routed: 27 · Census areas with a longer drive to a 2026 site: 3 · Cut off means: over 45 minutes or no route · Most minutes added to any known site: 92 · Most minutes added to the nearest 2026 site: 3.8 · Typical minutes added (median, where longer): 2.4 · People in areas cut off: 951 · People with a longer trip: 12,064 · People losing known sites within reach: 68,353 · Within reach means: 45-minute drive · Why this road:

   Sources: Esri ArcGIS routing with road closures (build time); highway centrelines: Hawai‘i DOT HPMS and surveyed mile markers (Hawai‘i Statewide GIS); Esri ArcGIS routing (drive-time matrix, normal traffic, build time); county announcements of sites opened in 2026; FEMA flood zones and tsunami evacuation zones (Hawai‘i Statewide GIS); Hawai‘i DOE public schools (Hawai‘i Statewide GIS); Hawai‘i Sea Level Rise Viewer, potential flooded highways at 3.2 ft (PacIOOS, via Hawai‘i Statewide GIS). Team assumptions: Future sea-level-rise scenario (3.2 ft; State of Hawai‘i SLR-XA), tested against 2026 sites: every potentially flooded highway segment on the island closed at once: all segments.

2. **Honoapi‘ilani Highway (Route 30) at the Pali and at Olowalu** (test 24: Road closures (driving))
   68,353 people in areas with a longer drive, fewer known sites within 45 minutes, or cut off (+103.7 min to any known site; +0 min to a 2026 site).

   Census areas cut off: none · Census areas cut off (over the limit or no route): 0 · Census areas cut off from every 2026 site: 0 · Census areas losing known sites within reach: 18 · Census areas with no route at all: 0 · Census areas routed: 41 · Census areas with a longer drive to a 2026 site: 0 · Cut off means: over 45 minutes or no route · Most minutes added to any known site: 103.7 · Most minutes added to the nearest 2026 site: 0 · Typical minutes added (median, where longer): 0 · People in areas cut off: 0 · People with a longer trip: 0 · People losing known sites within reach: 68,353 · Within reach means: 45-minute drive · Why this road: Only coastal highway between West Maui and Central Maui; the Pali stretch is exposed to rockfall and the Olowalu stretch to wildfire and coastal flooding (team assumption).

   Sources: Esri ArcGIS routing with road closures (build time); highway centrelines: Hawai‘i DOT HPMS and surveyed mile markers (Hawai‘i Statewide GIS); Esri ArcGIS routing (drive-time matrix, normal traffic, build time); county announcements of sites opened in 2026; FEMA flood zones and tsunami evacuation zones (Hawai‘i Statewide GIS); Hawai‘i DOE public schools (Hawai‘i Statewide GIS). Team assumptions: Closure location along the highway (a stretch of about 0.8 km): team assumption.

3. **Stranded visitors for 3 or 7 days – Maui** (test 21: Stranded visitors)
   55,290 visitors on an average day (island level).
   By length of stay: 3 days: 331,740 meals, 165,870 gallons of water; 7 days: 774,060 meals, 387,030 gallons of water · Visitors on an average day (2025): 55,290
   Sources: DBEDT State of Hawai‘i Data Book 2025, Table 7.06 (average daily visitor census, 2025); Florida Division of Emergency Management, Sheltering Guidance Aid (2010). Team assumptions: Every visitor on the island is stranded and needs meals and water from mass care (upper bound): 100%.
4. **All four counties at once (inland flooding)** (test 9: Several counties at once)
   25,452 people who may need a public shelter (likely), counties together (cots: 25,452; shelter staff per shift: 1,723).
   Cots: 25,452 · Counties: Kauai, Honolulu, Maui, Hawaii · People with limited English in shelter, by language (interpreters): Chinese: 100; Chuukese: 24; Hawaiian: 12; Ilocano: 153; Japanese: 82; Korean: 40; Marshallese: 10; Pidgin (Hawai‘i Creole English): 12; Spanish: 13; Tagalog/Filipino: 124; Vietnamese: 27 · Meals per day: 50,904 · People in shelter (likely): 25,452 · Rounding: counties' unrounded numbers added, then rounded up once · Shelter staff per shift: 1,723 · Water (gallons per day): 25,452
   Sources: CDC/ATSDR Social Vulnerability Index 2022 (ACS 2018–2022), HHS emPOWER, FEMA National Risk Index; 2020 Census blocks; Florida Division of Emergency Management, Sheltering Guidance Aid (2010); ACS PUMS 2019–2023 5-year (limited-English speakers by language); shelter-seeking rates in the plan (South Carolina EMD 2026 Annex G; Maui 2023 observed; team assumptions). Team assumptions: One rounding rule for counties together: add the unrounded county numbers, then round up once: round up.
5. **Honolulu and Maui at once (inland flooding)** (test 9: Several counties at once)
   21,106 people who may need a public shelter (likely), counties together (cots: 21,106; shelter staff per shift: 1,423).
   Cots: 21,106 · Counties: Honolulu, Maui · People with limited English in shelter, by language (interpreters): Chinese: 97; Chuukese: 24; Hawaiian: 8; Ilocano: 123; Japanese: 73; Korean: 40; Marshallese: 7; Spanish: 5; Tagalog/Filipino: 113; Vietnamese: 27 · Meals per day: 42,212 · People in shelter (likely): 21,106 · Rounding: counties' unrounded numbers added, then rounded up once · Shelter staff per shift: 1,423 · Water (gallons per day): 21,106
   Sources: CDC/ATSDR Social Vulnerability Index 2022 (ACS 2018–2022), HHS emPOWER, FEMA National Risk Index; 2020 Census blocks; Florida Division of Emergency Management, Sheltering Guidance Aid (2010); ACS PUMS 2019–2023 5-year (limited-English speakers by language); shelter-seeking rates in the plan (South Carolina EMD 2026 Annex G; Maui 2023 observed; team assumptions). Team assumptions: One rounding rule for counties together: add the unrounded county numbers, then round up once: round up.

6. **Lose Mayor Hannibal Tavares Community Center (Social Hall) and South Maui Community Park Gym (inland flooding)** (test 2: Two sites lost)
   16,851 people in the risk area pushed outside a 15-minute drive to a usable 2026 site (Census areas pushed outside a 15-minute drive: 4; people more than losing either site alone: 6,552).
   Census areas left with no usable 2026 site on the island: 0 · Census areas pushed outside a 15-minute drive: 4 · Drive limit (minutes): 15 · Island: Maui · People more than losing either site alone: 6,552 · People left with no usable 2026 site on the island: 0 · Sites: Mayor Hannibal Tavares Community Center (Social Hall), South Maui Community Park Gym
   Sources: CDC/ATSDR Social Vulnerability Index 2022 (ACS 2018–2022), HHS emPOWER, FEMA National Risk Index; 2020 Census blocks; county announcements of sites opened in 2026; FEMA flood zones and tsunami evacuation zones (Hawai‘i Statewide GIS); Esri ArcGIS routing (drive-time matrix, normal traffic, build time).
7. **Lose Baldwin High School and Mayor Hannibal Tavares Community Center (Social Hall) (inland flooding)** (test 2: Two sites lost)
   15,651 people in the risk area pushed outside a 15-minute drive to a usable 2026 site (Census areas pushed outside a 15-minute drive: 3; people more than losing either site alone: 9,099).
   Census areas left with no usable 2026 site on the island: 0 · Census areas pushed outside a 15-minute drive: 3 · Drive limit (minutes): 15 · Island: Maui · People more than losing either site alone: 9,099 · People left with no usable 2026 site on the island: 0 · Sites: Baldwin High School, Mayor Hannibal Tavares Community Center (Social Hall)
   Sources: CDC/ATSDR Social Vulnerability Index 2022 (ACS 2018–2022), HHS emPOWER, FEMA National Risk Index; 2020 Census blocks; county announcements of sites opened in 2026; FEMA flood zones and tsunami evacuation zones (Hawai‘i Statewide GIS); Esri ArcGIS routing (drive-time matrix, normal traffic, build time).
8. **Honoapi‘ilani Highway (Route 30) at the Pali and at Olowalu – walking** (test 25: Road closures (walking))
   12,413 people in areas with a longer walk to a 2026 site or cut off on foot (+627.6 min to any known site; +627.6 min to a 2026 site).
   Census areas cut off: none · Census areas cut off (over the limit or no route): 0 · Census areas cut off from every 2026 site: 0 · Census areas losing known sites within reach: 0 · Census areas with no route at all: 0 · Census areas routed: 39 · Census areas with a longer drive to a 2026 site: 3 · Cut off means: no walking route · Most minutes added to any known site: 627.6 · Most minutes added to the nearest 2026 site: 627.6 · Typical minutes added (median, where longer): 12.9 · People in areas cut off: 0 · People with a longer trip: 12,413 · People losing known sites within reach: 0 · Within reach means: walking route (sites within 12 km) · Why this road: Only coastal highway between West Maui and Central Maui; the Pali stretch is exposed to rockfall and the Olowalu stretch to wildfire and coastal flooding (team assumption).
   Sources: Esri ArcGIS routing with road closures (build time); highway centrelines: Hawai‘i DOT HPMS and surveyed mile markers (Hawai‘i Statewide GIS); Esri ArcGIS routing (walking time, build time); county announcements of sites opened in 2026; FEMA flood zones and tsunami evacuation zones (Hawai‘i Statewide GIS); Hawai‘i DOE public schools (Hawai‘i Statewide GIS). Team assumptions: Closure location along the highway (a stretch of about 0.8 km): team assumption.

9. **Hāna Highway (Route 360) near Keʻanae – walking** (test 25: Road closures (walking))
   11,462 people in areas with a longer walk to a 2026 site or cut off on foot (+12.9 min to any known site; +12.9 min to a 2026 site).

   Census areas cut off: none · Census areas cut off (over the limit or no route): 0 · Census areas cut off from every 2026 site: 0 · Census areas losing known sites within reach: 0 · Census areas with no route at all: 0 · Census areas routed: 39 · Census areas with a longer drive to a 2026 site: 2 · Cut off means: no walking route · Most minutes added to any known site: 12.9 · Most minutes added to the nearest 2026 site: 12.9 · Typical minutes added (median, where longer): 12.6 · People in areas cut off: 0 · People with a longer trip: 11,462 · People losing known sites within reach: 0 · Within reach means: walking route (sites within 12 km) · Why this road: Narrow road with many one-lane bridges and landslide-prone cliffs (team assumption).

   Sources: Esri ArcGIS routing with road closures (build time); highway centrelines: Hawai‘i DOT HPMS and surveyed mile markers (Hawai‘i Statewide GIS); Esri ArcGIS routing (walking time, build time); county announcements of sites opened in 2026; FEMA flood zones and tsunami evacuation zones (Hawai‘i Statewide GIS); Hawai‘i DOE public schools (Hawai‘i Statewide GIS). Team assumptions: Closure location along the highway (a stretch of about 0.8 km): team assumption.

10. **Lose South Maui Community Park Gym (inland flooding)** (test 1: One site lost)
    10,299 people in the risk area pushed outside a 15-minute drive to a usable 2026 site (Census areas pushed outside a 15-minute drive: 3).

    Census areas pushed outside a 15-minute drive: 3 · Drive limit (minutes): 15 · Island: Maui · May need a public shelter (likely): 515 · Site: South Maui Community Park Gym

    Sources: CDC/ATSDR Social Vulnerability Index 2022 (ACS 2018–2022), HHS emPOWER, FEMA National Risk Index; 2020 Census blocks; county announcements of sites opened in 2026; FEMA flood zones and tsunami evacuation zones (Hawai‘i Statewide GIS); Esri ArcGIS routing (drive-time matrix, normal traffic, build time).

Coastal flooding: top 10

1. **Future sea-level-rise scenario (3.2 ft; State of Hawai‘i SLR-XA), tested against 2026 sites – Maui: every potentially flooded highway segment closed** (test 24: Road closures (driving))
   70,875 people in areas with a longer drive, fewer known sites within 45 minutes, or cut off (+92 min to any known site; +3.8 min to a 2026 site; cut off from every 2026 site: Launiupoko / Mā‘alaea · Census area 320).

   Census areas cut off: Launiupoko / Mā‘alaea · Census area 320 · Census areas cut off (over the limit or no route): 1 · Census areas cut off from every 2026 site: 1 · Census areas losing known sites within reach: 18 · Census areas with no route at all: 1 · Census areas routed: 27 · Census areas with a longer drive to a 2026 site: 3 · Cut off means: over 45 minutes or no route · Most minutes added to any known site: 92 · Most minutes added to the nearest 2026 site: 3.8 · Typical minutes added (median, where longer): 2.4 · People in areas cut off: 951 · People with a longer trip: 12,064 · People losing known sites within reach: 68,353 · Within reach means: 45-minute drive · Why this road:

   Sources: Esri ArcGIS routing with road closures (build time); highway centrelines: Hawai‘i DOT HPMS and surveyed mile markers (Hawai‘i Statewide GIS); Esri ArcGIS routing (drive-time matrix, normal traffic, build time); county announcements of sites opened in 2026; FEMA flood zones and tsunami evacuation zones (Hawai‘i Statewide GIS); Hawai‘i DOE public schools (Hawai‘i Statewide GIS); Hawai‘i Sea Level Rise Viewer, potential flooded highways at 3.2 ft (PacIOOS, via Hawai‘i Statewide GIS). Team assumptions: Future sea-level-rise scenario (3.2 ft; State of Hawai‘i SLR-XA), tested against 2026 sites: every potentially flooded highway segment on the island closed at once: all segments.

2. **Honoapi‘ilani Highway (Route 30) at the Pali and at Olowalu** (test 24: Road closures (driving))
   68,353 people in areas with a longer drive, fewer known sites within 45 minutes, or cut off (+103.7 min to any known site; +0 min to a 2026 site).

   Census areas cut off: none · Census areas cut off (over the limit or no route): 0 · Census areas cut off from every 2026 site: 0 · Census areas losing known sites within reach: 18 · Census areas with no route at all: 0 · Census areas routed: 41 · Census areas with a longer drive to a 2026 site: 0 · Cut off means: over 45 minutes or no route · Most minutes added to any known site: 103.7 · Most minutes added to the nearest 2026 site: 0 · Typical minutes added (median, where longer): 0 · People in areas cut off: 0 · People with a longer trip: 0 · People losing known sites within reach: 68,353 · Within reach means: 45-minute drive · Why this road: Only coastal highway between West Maui and Central Maui; the Pali stretch is exposed to rockfall and the Olowalu stretch to wildfire and coastal flooding (team assumption).

   Sources: Esri ArcGIS routing with road closures (build time); highway centrelines: Hawai‘i DOT HPMS and surveyed mile markers (Hawai‘i Statewide GIS); Esri ArcGIS routing (drive-time matrix, normal traffic, build time); county announcements of sites opened in 2026; FEMA flood zones and tsunami evacuation zones (Hawai‘i Statewide GIS); Hawai‘i DOE public schools (Hawai‘i Statewide GIS). Team assumptions: Closure location along the highway (a stretch of about 0.8 km): team assumption.

3. **Stranded visitors for 3 or 7 days – Maui** (test 21: Stranded visitors)
   55,290 visitors on an average day (island level).
   By length of stay: 3 days: 331,740 meals, 165,870 gallons of water; 7 days: 774,060 meals, 387,030 gallons of water · Visitors on an average day (2025): 55,290
   Sources: DBEDT State of Hawai‘i Data Book 2025, Table 7.06 (average daily visitor census, 2025); Florida Division of Emergency Management, Sheltering Guidance Aid (2010). Team assumptions: Every visitor on the island is stranded and needs meals and water from mass care (upper bound): 100%.
4. **All four counties at once (coastal flooding)** (test 9: Several counties at once)
   17,662 people who may need a public shelter (likely), counties together (cots: 17,662; shelter staff per shift: 1,171).
   Cots: 17,662 · Counties: Kauai, Honolulu, Maui, Hawaii · People with limited English in shelter, by language (interpreters): Chinese: 99; Chuukese: 23; Hawaiian: 13; Ilocano: 154; Japanese: 83; Korean: 39; Marshallese: 11; Pidgin (Hawai‘i Creole English): 15; Spanish: 15; Tagalog/Filipino: 122; Vietnamese: 26 · Meals per day: 35,324 · People in shelter (likely): 17,662 · Rounding: counties' unrounded numbers added, then rounded up once · Shelter staff per shift: 1,171 · Water (gallons per day): 17,662
   Sources: CDC/ATSDR Social Vulnerability Index 2022 (ACS 2018–2022), HHS emPOWER, FEMA National Risk Index; 2020 Census blocks; Florida Division of Emergency Management, Sheltering Guidance Aid (2010); ACS PUMS 2019–2023 5-year (limited-English speakers by language); shelter-seeking rates in the plan (South Carolina EMD 2026 Annex G; Maui 2023 observed; team assumptions). Team assumptions: One rounding rule for counties together: add the unrounded county numbers, then round up once: round up.
5. **Honolulu and Maui at once (coastal flooding)** (test 9: Several counties at once)
   14,005 people who may need a public shelter (likely), counties together (cots: 14,005; shelter staff per shift: 926).
   Cots: 14,005 · Counties: Honolulu, Maui · People with limited English in shelter, by language (interpreters): Chinese: 94; Chuukese: 23; Hawaiian: 8; Ilocano: 121; Japanese: 71; Korean: 39; Marshallese: 7; Spanish: 5; Tagalog/Filipino: 110; Vietnamese: 26 · Meals per day: 28,009 · People in shelter (likely): 14,005 · Rounding: counties' unrounded numbers added, then rounded up once · Shelter staff per shift: 926 · Water (gallons per day): 14,005
   Sources: CDC/ATSDR Social Vulnerability Index 2022 (ACS 2018–2022), HHS emPOWER, FEMA National Risk Index; 2020 Census blocks; Florida Division of Emergency Management, Sheltering Guidance Aid (2010); ACS PUMS 2019–2023 5-year (limited-English speakers by language); shelter-seeking rates in the plan (South Carolina EMD 2026 Annex G; Maui 2023 observed; team assumptions). Team assumptions: One rounding rule for counties together: add the unrounded county numbers, then round up once: round up.

6. **Lose Lahaina Civic Center Gym and South Maui Community Park Gym (coastal flooding)** (test 2: Two sites lost)
   12,611 people in the risk area pushed outside a 15-minute drive to a usable 2026 site (Census areas pushed outside a 15-minute drive: 4; people more than losing either site alone: 3,100).
   Census areas left with no usable 2026 site on the island: 0 · Census areas pushed outside a 15-minute drive: 4 · Drive limit (minutes): 15 · Island: Maui · People more than losing either site alone: 3,100 · People left with no usable 2026 site on the island: 0 · Sites: Lahaina Civic Center Gym, South Maui Community Park Gym
   Sources: CDC/ATSDR Social Vulnerability Index 2022 (ACS 2018–2022), HHS emPOWER, FEMA National Risk Index; 2020 Census blocks; county announcements of sites opened in 2026; FEMA flood zones and tsunami evacuation zones (Hawai‘i Statewide GIS); Esri ArcGIS routing (drive-time matrix, normal traffic, build time).

7. **Honoapi‘ilani Highway (Route 30) at the Pali and at Olowalu – walking** (test 25: Road closures (walking))
   12,413 people in areas with a longer walk to a 2026 site or cut off on foot (+627.6 min to any known site; +627.6 min to a 2026 site).
   Census areas cut off: none · Census areas cut off (over the limit or no route): 0 · Census areas cut off from every 2026 site: 0 · Census areas losing known sites within reach: 0 · Census areas with no route at all: 0 · Census areas routed: 39 · Census areas with a longer drive to a 2026 site: 3 · Cut off means: no walking route · Most minutes added to any known site: 627.6 · Most minutes added to the nearest 2026 site: 627.6 · Typical minutes added (median, where longer): 12.9 · People in areas cut off: 0 · People with a longer trip: 12,413 · People losing known sites within reach: 0 · Within reach means: walking route (sites within 12 km) · Why this road: Only coastal highway between West Maui and Central Maui; the Pali stretch is exposed to rockfall and the Olowalu stretch to wildfire and coastal flooding (team assumption).
   Sources: Esri ArcGIS routing with road closures (build time); highway centrelines: Hawai‘i DOT HPMS and surveyed mile markers (Hawai‘i Statewide GIS); Esri ArcGIS routing (walking time, build time); county announcements of sites opened in 2026; FEMA flood zones and tsunami evacuation zones (Hawai‘i Statewide GIS); Hawai‘i DOE public schools (Hawai‘i Statewide GIS). Team assumptions: Closure location along the highway (a stretch of about 0.8 km): team assumption.

8. **Hāna Highway (Route 360) near Ke‘anae – walking** (test 25: Road closures (walking))
   11,462 people in areas with a longer walk to a 2026 site or cut off on foot (+12.9 min to any known site; +12.9 min to a 2026 site).
   Census areas cut off: none · Census areas cut off (over the limit or no route): 0 · Census areas cut off from every 2026 site: 0 · Census areas losing known sites within reach: 0 · Census areas with no route at all: 0 · Census areas routed: 39 · Census areas with a longer drive to a 2026 site: 2 · Cut off means: no walking route · Most minutes added to any known site: 12.9 · Most minutes added to the nearest 2026 site: 12.9 · Typical minutes added (median, where longer): 12.6 · People in areas cut off: 0 · People with a longer trip: 11,462 · People losing known sites within reach: 0 · Within reach means: walking route (sites within 12 km) · Why this road: Narrow road with many one-lane bridges and landslide-prone cliffs (team assumption).
   Sources: Esri ArcGIS routing with road closures (build time); highway centrelines: Hawai‘i DOT HPMS and surveyed mile markers (Hawai‘i Statewide GIS); Esri ArcGIS routing (walking time, build time); county announcements of sites opened in 2026; FEMA flood zones and tsunami evacuation zones (Hawai‘i Statewide GIS); Hawai‘i DOE public schools (Hawai‘i Statewide GIS). Team assumptions: Closure location along the highway (a stretch of about 0.8 km): team assumption.

9. **Lose South Maui Community Park Gym (coastal flooding)** (test 1: One site lost)
   9,511 people in the risk area pushed outside a 15-minute drive to a usable 2026 site (Census areas pushed outside a 15-minute drive: 3).
   Census areas pushed outside a 15-minute drive: 3 · Drive limit (minutes): 15 · Island: Maui · May need a public shelter (likely): 476 · Site: South Maui Community Park Gym
   Sources: CDC/ATSDR Social Vulnerability Index 2022 (ACS 2018–2022), HHS emPOWER, FEMA National Risk Index; 2020 Census blocks; county announcements of sites opened in 2026; FEMA flood zones and tsunami evacuation zones (Hawai‘i Statewide GIS); Esri ArcGIS routing (drive-time matrix, normal traffic, build time).
10. **Lose Baldwin High School and South Maui Community Park Gym (coastal flooding)** (test 2: Two sites lost)
    9,511 people in the risk area pushed outside a 15-minute drive to a usable 2026 site (Census areas pushed outside a 15-minute drive: 3; people more than losing either site alone: 0).
    Census areas left with no usable 2026 site on the island: 0 · Census areas pushed outside a 15-minute drive: 3 · Drive limit (minutes): 15 · Island: Maui · People more than losing either site alone: 0 · People left with no usable 2026 site on the island: 0 · Sites: Baldwin High School, South Maui Community Park Gym
    Sources: CDC/ATSDR Social Vulnerability Index 2022 (ACS 2018–2022), HHS emPOWER, FEMA National Risk Index; 2020 Census blocks; county announcements of sites opened in 2026; FEMA flood zones and tsunami evacuation zones (Hawai‘i Statewide GIS); Esri ArcGIS routing (drive-time matrix, normal traffic, build time).

Every result, ranked by people affected

Rank	Test	Scenario	Hazard	Result
1	24	Future sea-level-rise scenario (3.2 ft; State of Hawai‘i SLR-XA), tested against 2026 sites – Maui: every potentially flooded highway segment closed	All hazards	70,875 people in areas with a longer drive, fewer known sites within 45 minutes, or cut off (+92 min to any known site; +3.8 min to a 2026 site; cut off from every 2026 site: Launiupoko / Mā‘alaea · Census area 320)
1	9	All four counties at once (hurricane)	Hurricane	43,518 people who may need a public shelter (likely), counties together (cots: 43,518; shelter staff per shift: 2,921)
2	2	Lose Baldwin High School and Mayor Hannibal Tavares Community Center (Social Hall) (hurricane)	Hurricane	69,957 people in the risk area pushed outside a 15-minute drive to a usable 2026 site (Census areas pushed outside a 15-minute drive: 15; people more than losing either site alone: 23,964)
3	2	Lose Baldwin High School and South Maui Community Park Gym (hurricane)	Hurricane	68,514 people in the risk area pushed outside a 15-minute drive to a usable 2026 site (Census areas pushed outside a 15-minute drive: 18; people more than losing either site alone: 22,521)
3	9	Honolulu and Maui at once (hurricane)	Hurricane	35,248 people who may need a public shelter (likely), counties together (cots: 35,248; shelter staff per shift: 2,357)
4	24	Honoapi‘ilani Highway (Route 30) at the Pali and at Olowalu	All hazards	68,353 people in areas with a longer drive, fewer known sites within 45 minutes, or cut off (+103.7 min to any known site; +0 min to a 2026 site)
5	23b	Areas high in the State Fire Risk layer but not in the plan's wildfire areas, and the reverse – Maui	Wildfire	56,637 people in areas rated high only by the State Fire Risk layer (Census areas high in the State layer only: 18; Census areas in the plan's wildfire areas only: 4)

Rank	Test	Scenario	Hazard	Result
5	9	All four counties at once (inland flooding)	Inland flooding	25,452 people who may need a public shelter (likely), counties together (cots: 25,452; shelter staff per shift: 1,723)
6	20	Visitors on the island(s) – Maui (tsunami)	Tsunami	55,290 visitors on an average day (island level) (visitors inside the tsunami zone: needs county data (hotel and rental locations))
6	9	All four counties at once (wildfire)	Wildfire	22,115 people who may need a public shelter (likely), counties together (cots: 22,115; shelter staff per shift: 1,534)
7	21	Stranded visitors for 3 or 7 days – Maui	All hazards	55,290 visitors on an average day (island level)
8	2	Lose Baldwin High School and Lahaina Civic Center Gym (hurricane)	Hurricane	49,093 people in the risk area pushed outside a 15-minute drive to a usable 2026 site (Census areas pushed outside a 15-minute drive: 11; people more than losing either site alone: 3,100)
8	9	Honolulu and Maui at once (inland flooding)	Inland flooding	21,106 people who may need a public shelter (likely), counties together (cots: 21,106; shelter staff per shift: 1,423)
9	1	Lose Baldwin High School (hurricane)	Hurricane	45,993 people in the risk area pushed outside a 15-minute drive to a usable 2026 site (Census areas pushed outside a 15-minute drive: 10)
11	2	Lose Mayor Hannibal Tavares Community Center (Social Hall) and South Maui Community Park Gym (hurricane)	Hurricane	41,160 people in the risk area pushed outside a 15-minute drive to a usable 2026 site (Census areas pushed outside a 15-minute drive: 12; people more than losing either site alone: 18,639)
11	9	Honolulu and Maui at once (wildfire)	Wildfire	18,155 people who may need a public shelter (likely), counties together (cots: 18,155; shelter staff per shift: 1,258)

Rank	Test	Scenario	Hazard	Result
12	8	Extreme tsunami zone (worst case) – Maui	Tsunami	38,773 people in the extreme tsunami evacuation zone (may need a public shelter (likely): 1,939; 2026 sites inside the extreme zone: 0)
12	9	All four counties at once (coastal flooding)	Coastal flooding	17,662 people who may need a public shelter (likely), counties together (cots: 17,662; shelter staff per shift: 1,171)
13	2	Lose Lahaina Civic Center Gym and South Maui Community Park Gym (hurricane)	Hurricane	25,621 people in the risk area pushed outside a 15-minute drive to a usable 2026 site (Census areas pushed outside a 15-minute drive: 9; people more than losing either site alone: 3,100)
14	1	Lose South Maui Community Park Gym (hurricane)	Hurricane	22,521 people in the risk area pushed outside a 15-minute drive to a usable 2026 site (Census areas pushed outside a 15-minute drive: 8)
15	9	Honolulu and Maui at once (coastal flooding)	Coastal flooding	14,005 people who may need a public shelter (likely), counties together (cots: 14,005; shelter staff per shift: 926)
16	2	Lose Lahaina Civic Center Gym and Mayor Hannibal Tavares Community Center (Social Hall) (hurricane)	Hurricane	21,739 people in the risk area pushed outside a 15-minute drive to a usable 2026 site (Census areas pushed outside a 15-minute drive: 5; people more than losing either site alone: 3,100)
17	1	Lose Mayor Hannibal Tavares Community Center (Social Hall) (hurricane)	Hurricane	18,639 people in the risk area pushed outside a 15-minute drive to a usable 2026 site (Census areas pushed outside a 15-minute drive: 4)
17	9	Maui and Hawai‘i at once (hurricane)	Hurricane	11,010 people who may need a public shelter (likely), counties together (cots: 11,010; shelter staff per shift: 744)
19	2	Lose Mayor Hannibal Tavares Community Center (Social Hall) and South Maui Community Park Gym (inland flooding)	Inland flooding	16,851 people in the risk area pushed outside a 15-minute drive to a usable 2026 site (Census areas pushed outside a 15-minute drive: 4; people more than losing either site alone: 6,552)

Rank	Test	Scenario	Hazard	Result
19	9	Kauaʻi and Maui at once (hurricane)	Hurricane	7,150 people who may need a public shelter (likely), counties together (cots: 7,150; shelter staff per shift: 488)
20	2	Lose Baldwin High School and Mayor Hannibal Tavares Community Center (Social Hall) (inland flooding)	Inland flooding	15,651 people in the risk area pushed outside a 15-minute drive to a usable 2026 site (Census areas pushed outside a 15-minute drive: 3; people more than losing either site alone: 9,099)
20	9	All four counties at once (tsunami)	Tsunami	6,382 people who may need a public shelter (likely), counties together (cots: 6,382; shelter staff per shift: 442)
21	2	Lose Baldwin High School and South Maui Community Park Gym (inland flooding)	Inland flooding	14,073 people in the risk area pushed outside a 15-minute drive to a usable 2026 site (Census areas pushed outside a 15-minute drive: 4; people more than losing either site alone: 3,774)
21	9	Maui and Hawaiʻi at once (inland flooding)	Inland flooding	5,909 people who may need a public shelter (likely), counties together (cots: 5,909; shelter staff per shift: 401)
22	2	Lose Lahaina Civic Center Gym and South Maui Community Park Gym (coastal flooding)	Coastal flooding	12,611 people in the risk area pushed outside a 15-minute drive to a usable 2026 site (Census areas pushed outside a 15-minute drive: 4; people more than losing either site alone: 3,100)
22	9	Honolulu and Maui at once (tsunami)	Tsunami	5,326 people who may need a public shelter (likely), counties together (cots: 5,326; shelter staff per shift: 362)
23	25	Honoapiʻilani Highway (Route 30) at the Pali and at Olowalu – walking	All hazards	12,413 people in areas with a longer walk to a 2026 site or cut off on foot (+627.6 min to any known site; +627.6 min to a 2026 site)
23	9	Maui and Hawaiʻi at once (wildfire)	Wildfire	5,320 people who may need a public shelter (likely), counties together (cots: 5,320; shelter staff per shift: 371)

Rank	Test	Scenario	Hazard	Result
24	2	Lose Baldwin High School and Mayor Hannibal Tavares Community Center (Social Hall) (wildfire)	Wildfire	12,302 people in the risk area pushed outside a 15-minute drive to a usable 2026 site (Census areas pushed outside a 15-minute drive: 2; people more than losing either site alone: 5,325)
25	2	Lose Lahaina Civic Center Gym and South Maui Community Park Gym (tsunami)	Tsunami	12,189 people in the risk area pushed outside a 15-minute drive to a usable 2026 site (Census areas pushed outside a 15-minute drive: 8; people more than losing either site alone: 3,072)
25	9	Maui and Hawai‘i at once (coastal flooding)	Coastal flooding	4,533 people who may need a public shelter (likely), counties together (cots: 4,533; shelter staff per shift: 303)
26	25	Hāna Highway (Route 360) near Ke‘anae – walking	All hazards	11,462 people in areas with a longer walk to a 2026 site or cut off on foot (+12.9 min to any known site; +12.9 min to a 2026 site)
27	2	Lose Baldwin High School and South Maui Community Park Gym (tsunami)	Tsunami	11,323 people in the risk area pushed outside a 15-minute drive to a usable 2026 site (Census areas pushed outside a 15-minute drive: 10; people more than losing either site alone: 2,206)
28	2	Lose Baldwin High School and South Maui Community Park Gym (wildfire)	Wildfire	10,561 people in the risk area pushed outside a 15-minute drive to a usable 2026 site (Census areas pushed outside a 15-minute drive: 2; people more than losing either site alone: 3,584)
29	1	Lose South Maui Community Park Gym (inland flooding)	Inland flooding	10,299 people in the risk area pushed outside a 15-minute drive to a usable 2026 site (Census areas pushed outside a 15-minute drive: 3)
29	9	Kaua‘i and Maui at once (inland flooding)	Inland flooding	3,703 people who may need a public shelter (likely), counties together (cots: 3,703; shelter staff per shift: 255)

Rank	Test	Scenario	Hazard	Result
31	9	Kauaʻi and Maui at once (wildfire)	Wildfire	3,524 people who may need a public shelter (likely), counties together (cots: 3,524; shelter staff per shift: 249)
32	1	Lose South Maui Community Park Gym (coastal flooding)	Coastal flooding	9,511 people in the risk area pushed outside a 15-minute drive to a usable 2026 site (Census areas pushed outside a 15-minute drive: 3)
32	9	Kauaʻi and Maui at once (coastal flooding)	Coastal flooding	2,794 people who may need a public shelter (likely), counties together (cots: 2,794; shelter staff per shift: 194)
33	9	Maui and Hawaiʻi at once (tsunami)	Tsunami	1,914 people who may need a public shelter (likely), counties together (cots: 1,914; shelter staff per shift: 136)
34	9	Kauaʻi and Maui at once (tsunami)	Tsunami	1,760 people who may need a public shelter (likely), counties together (cots: 1,760; shelter staff per shift: 126)
36	1	Lose South Maui Community Park Gym (tsunami)	Tsunami	9,117 people in the risk area pushed outside a 15-minute drive to a usable 2026 site (Census areas pushed outside a 15-minute drive: 7)
39	1	Lose Baldwin High School (wildfire)	Wildfire	6,977 people in the risk area pushed outside a 15-minute drive to a usable 2026 site (Census areas pushed outside a 15-minute drive: 1)
42	1	Lose Mayor Hannibal Tavares Community Center (Social Hall) (inland flooding)	Inland flooding	6,552 people in the risk area pushed outside a 15-minute drive to a usable 2026 site (Census areas pushed outside a 15-minute drive: 1)
45	5	Hurricane and inland flooding together – Maui	Hurricane + Inland flooding	5,997 people who may need a public shelter (likely) (2026 sites usable for every hazard in the pair: 7; site gap: 17)
46	2	Lose Baldwin High School and Mayor Hannibal Tavares Community Center (Social Hall) (coastal flooding)	Coastal flooding	5,325 people in the risk area pushed outside a 15-minute drive to a usable 2026 site (Census areas pushed outside a 15-minute drive: 1; people more than losing either site alone: 5,325)

Rank	Test	Scenario	Hazard	Result
47	2	Lose Baldwin High School and Lahaina Civic Center Gym (tsunami)	Tsunami	5,278 people in the risk area pushed outside a 15-minute drive to a usable 2026 site (Census areas pushed outside a 15-minute drive: 4; people more than losing either site alone: 2,206)
48	11	Stays of 3 to 30 days – Maui (hurricane)	Hurricane	4,944 people who may need a public shelter (likely)
49	2	Lose Baldwin High School and Mayor Hannibal Tavares Community Center (Social Hall) (tsunami)	Tsunami	4,800 people in the risk area pushed outside a 15-minute drive to a usable 2026 site (Census areas pushed outside a 15-minute drive: 4; people more than losing either site alone: 2,593)
50	1	Lose Baldwin High School (inland flooding)	Inland flooding	3,774 people in the risk area pushed outside a 15-minute drive to a usable 2026 site (Census areas pushed outside a 15-minute drive: 1)
53	1	Lose South Maui Community Park Gym (wildfire)	Wildfire	3,584 people in the risk area pushed outside a 15-minute drive to a usable 2026 site (Census areas pushed outside a 15-minute drive: 1)
57	7	Dam failure and inland flooding together – Maui	Dam failure + Inland flooding	3,577 people who may need a public shelter (likely) (2026 sites usable for every hazard in the pair: 6; site gap: 9)
58	10	Rate at which demand needs more sites than are usable – Maui (hurricane)	Hurricane	3,296 people who may need a shelter at the rate where usable sites run out (rate when usable sites run out: 2%; usable 2026 sites: 8)
59	1	Lose Lānaʻi Youth Center (hurricane)	Hurricane	3,218 people in the risk area pushed outside a 15-minute drive to a usable 2026 site (Census areas pushed outside a 15-minute drive: 1)
60	1	Lose Lahaina Civic Center Gym (coastal flooding)	Coastal flooding	3,100 people in the risk area pushed outside a 15-minute drive to a usable 2026 site (Census areas pushed outside a 15-minute drive: 1)
61	1	Lose Lahaina Civic Center Gym (hurricane)	Hurricane	3,100 people in the risk area pushed outside a 15-minute drive to a usable 2026 site (Census areas pushed outside a 15-minute drive: 1)

Rank	Test	Scenario	Hazard	Result
66	1	Lose Lahaina Civic Center Gym (tsunami)	Tsunami	3,072 people in the risk area pushed outside a 15-minute drive to a usable 2026 site (Census areas pushed outside a 15-minute drive: 1)
69	1	Lose Molokaʻi Baptist Church (hurricane)	Hurricane	2,640 people in the risk area pushed outside a 15-minute drive to a usable 2026 site (Census areas pushed outside a 15-minute drive: 1)
70	1	Lose Molokaʻi Baptist Church (wildfire)	Wildfire	2,640 people in the risk area pushed outside a 15-minute drive to a usable 2026 site (Census areas pushed outside a 15-minute drive: 1)
71	2	Lose Kilohana Recreation Center and Molokaʻi Baptist Church (hurricane)	Hurricane	2,640 people in the risk area pushed outside a 15-minute drive to a usable 2026 site (Census areas pushed outside a 15-minute drive: 1; people more than losing either site alone: 0)
72	2	Lose Kilohana Recreation Center and Molokaʻi Baptist Church (wildfire)	Wildfire	2,640 people in the risk area pushed outside a 15-minute drive to a usable 2026 site (Census areas pushed outside a 15-minute drive: 1; people more than losing either site alone: 0)
73	11	Stays of 3 to 30 days – Maui (inland flooding)	Inland flooding	2,633 people who may need a public shelter (likely)
74	6	Wildfire and power loss together – Maui	Wildfire + Power loss	2,442 people who may need a public shelter (likely) (outlets per day if the power is out: 311)
75	10	Rate at which demand needs more sites than are usable – Maui (wildfire)	Wildfire	2,442 people who may need a shelter at the rate where usable sites run out (rate when usable sites run out: 5%; usable 2026 sites: 8)
76	11	Stays of 3 to 30 days – Maui (wildfire)	Wildfire	2,442 people who may need a public shelter (likely)
77	4	Tsunami and coastal flooding together – Maui	Tsunami + Coastal flooding	2,299 people who may need a public shelter (likely) (2026 sites usable for every hazard in the pair: 7; site gap: 3)
78	1	Lose Baldwin High School (tsunami)	Tsunami	2,206 people in the risk area pushed outside a 15-minute drive to a usable 2026 site (Census areas pushed outside a 15-minute drive: 3)

Rank	Test	Scenario	Hazard	Result
80	10	Rate at which demand needs more sites than are usable – Maui (inland flooding)	Inland flooding	2,106 people who may need a shelter at the rate where usable sites run out (rate when usable sites run out: 4%; usable 2026 sites: 7)
81	10	Rate at which demand needs more sites than are usable – Maui (coastal flooding)	Coastal flooding	1,834 people who may need a shelter at the rate where usable sites run out (rate when usable sites run out: 5%; usable 2026 sites: 7)
82	11	Stays of 3 to 30 days – Maui (coastal flooding)	Coastal flooding	1,834 people who may need a public shelter (likely)
83	10	Rate at which demand needs more sites than are usable – Maui (tsunami)	Tsunami	1,831 people who may need a shelter at the rate where usable sites run out (rate when usable sites run out: 7%; usable 2026 sites: 7)
84	11	Stays of 3 to 30 days – Maui (tsunami)	Tsunami	1,308 people who may need a public shelter (likely)
85	13	Kūpuna living alone, no car, far from a site – Maui (hurricane)	Hurricane	1,237 kūpuna living alone without a car, outside a 15-minute drive (upper bound) (Census areas outside a 15-minute drive: 20)
86	12	Power out 1 to 14 days – Maui (hurricane)	Hurricane	1,074 power-dependent residents in the risk area (outlets per day while the power is out: 1,074)
87	13	Kūpuna living alone, no car, far from a site – Maui (coastal flooding)	Coastal flooding	668 kūpuna living alone without a car, outside a 15-minute drive (upper bound) (Census areas outside a 15-minute drive: 7)
88	13	Kūpuna living alone, no car, far from a site – Maui (inland flooding)	Inland flooding	622 kūpuna living alone without a car, outside a 15-minute drive (upper bound) (Census areas outside a 15-minute drive: 6)
89	13	Kūpuna living alone, no car, far from a site – Maui (wildfire)	Wildfire	484 kūpuna living alone without a car, outside a 15-minute drive (upper bound) (Census areas outside a 15-minute drive: 8)
90	23a	No car, inside the tsunami zone, over 15 minutes' walk to a safe zone – Maui	Tsunami	463 people without a car more than 15 minutes' walk from a tsunami safe zone (Census areas over 15 minutes' walk: 3; longest walk (minutes): 16.8)

Rank	Test	Scenario	Hazard	Result
91	14	Hearing difficulty and outside siren range (tsunami, at night) – Maui	Tsunami	443 people with hearing difficulty outside rated siren range (upper bound)
92	15	Power-dependent and far from a site – Maui (hurricane)	Hurricane	402 power-dependent residents outside a 15-minute drive (Census areas outside a 15-minute drive: 20)
93	18	Add the 2026 Point-in-Time count – Maui (coastal flooding)	Coastal flooding	398 people added by the 2026 Point-in-Time count (increase over the likely demand: 21.7%)
94	18	Add the 2026 Point-in-Time count – Maui (hurricane)	Hurricane	398 people added by the 2026 Point-in-Time count (increase over the likely demand: 8.1%)
95	18	Add the 2026 Point-in-Time count – Maui (inland flooding)	Inland flooding	398 people added by the 2026 Point-in-Time count (increase over the likely demand: 15.1%)
96	18	Add the 2026 Point-in-Time count – Maui (tsunami)	Tsunami	398 people added by the 2026 Point-in-Time count (increase over the likely demand: 30.4%)
97	18	Add the 2026 Point-in-Time count – Maui (wildfire)	Wildfire	398 people added by the 2026 Point-in-Time count (increase over the likely demand: 16.3%)
98	12	Power out 1 to 14 days – Maui (inland flooding)	Inland flooding	374 power-dependent residents in the risk area (outlets per day while the power is out: 374)
99	13	Kūpuna living alone, no car, far from a site – Maui (tsunami)	Tsunami	359 kūpuna living alone without a car, outside a 15-minute drive (upper bound) (Census areas outside a 15-minute drive: 14)
100	12	Power out 1 to 14 days – Maui (wildfire)	Wildfire	311 power-dependent residents in the risk area (outlets per day while the power is out: 311)
101	12	Power out 1 to 14 days – Maui (coastal flooding)	Coastal flooding	237 power-dependent residents in the risk area (outlets per day while the power is out: 237)
102	15	Power-dependent and far from a site – Maui (inland flooding)	Inland flooding	198 power-dependent residents outside a 15-minute drive (Census areas outside a 15-minute drive: 6)
103	15	Power-dependent and far from a site – Maui (wildfire)	Wildfire	193 power-dependent residents outside a 15-minute drive (Census areas outside a 15-minute drive: 8)

Rank	Test	Scenario	Hazard	Result
104	12	Power out 1 to 14 days – Maui (tsunami)	Tsunami	160 power-dependent residents in the risk area (outlets per day while the power is out: 160)
105	15	Power-dependent and far from a site – Maui (coastal flooding)	Coastal flooding	142 power-dependent residents outside a 15-minute drive (Census areas outside a 15-minute drive: 7)
106	15	Power-dependent and far from a site – Maui (tsunami)	Tsunami	58 power-dependent residents outside a 15-minute drive (Census areas outside a 15-minute drive: 14)
140	17	Pet households vs pet-friendly sites – Maui (hurricane)	Hurricane	1,161 households with a dog or a cat in shelter (usable sites that took pets in 2026 (count): 5)
141	17	Pet households vs pet-friendly sites – Maui (inland flooding)	Inland flooding	659 households with a dog or a cat in shelter (usable sites that took pets in 2026 (count): 5)
142	17	Pet households vs pet-friendly sites – Maui (wildfire)	Wildfire	552 households with a dog or a cat in shelter (usable sites that took pets in 2026 (count): 5)
143	17	Pet households vs pet-friendly sites – Maui (coastal flooding)	Coastal flooding	451 households with a dog or a cat in shelter (usable sites that took pets in 2026 (count): 5)
144	17	Pet households vs pet-friendly sites – Maui (tsunami)	Tsunami	323 households with a dog or a cat in shelter (usable sites that took pets in 2026 (count): 5)
145	19	Childcare centres in the hurricane zone – Maui	Hurricane	62 childcare centres in the zone (childcare centres in the county: 62)
146	19	Childcare centres in the wildfire zone – Maui	Wildfire	36 childcare centres in the zone (childcare centres in the county: 62)
147	19	Childcare centres in the tsunami zone – Maui	Tsunami	15 childcare centres in the zone (childcare centres in the county: 62)
148	19	Childcare centres in the coastal flooding zone – Maui	Coastal flooding	4 childcare centres in the zone (childcare centres in the county: 62)
149	19	Childcare centres in the inland flooding zone – Maui	Inland flooding	4 childcare centres in the zone (childcare centres in the county: 62)
150	22	Care facilities in the tsunami zone – Maui	Tsunami	3 care facilities in the zone (facilities over 15 minutes from a usable site, or none on the island: 3)
151	22	Care facilities in the coastal flooding zone – Maui	Coastal flooding	1 care facilities in the zone (facilities over 15 minutes from a usable site, or none on the island: 1)

Rank	Test	Scenario	Hazard	Result
152	22	Care facilities in the inland flooding zone – Maui	Inland flooding	1 care facilities in the zone (facilities over 15 minutes from a usable site, or none on the island: 1)
153	16	Chuukese: 0 of 15 messages translated – Maui	All hazards	people who speak English less than very well: needs county data (messages translated: 0)
154	16	Marshallese: 0 of 15 messages translated – Maui	All hazards	people who speak English less than very well: needs county data (messages translated: 0)

Questions for counties

These need county knowledge. The plan has no data for them, so it shows no numbers.

1. **Shelter buildings in an earthquake or a strong hurricane:** Which shelter buildings have been assessed for earthquake and hurricane-wind resilience, and which need a check before they are used after a strong event?
2. **Resupply for Molokaʻi and Lānaʻi:** How would water, food, cots, fuel and medical supplies reach Molokaʻi and Lānaʻi if the harbours or airports close for several days, and who decides the order of shipments?
3. **Cell network outages:** Which areas lose mobile phone service first in a long power outage or after wind damage, and how will warnings and shelter information reach them (radio, sirens, door-to-door)?
4. **Schools used as shelters:** When a school is used as a shelter, how long can it stay open before classes must restart, and where do residents go next?
5. **Partner capacity:** Could each county collect partner capacity (cots, meals, staff, vehicles, interpreters) with a county-run form such as Survey123, with no personal data, so the plan can show pledged capacity instead of 'confirm with partner'?